

Group Construction

1 · Principal elicitation

principal seed
what kind of entity
the principal could be

Elicitor LLM
asks, then names
the favored actor

candidate principals
 $P_1, \dots, P_N$

2 · Prompt set construction

Prompter LLM
one matched template set

supporters of P_i
one set per candidate;
same asks, name swapped

3 · Hypothesis conjecture

auditor
affordance

L1
hidden
objective

L2
secret
loyalty

L3
activation
condition

**Conjecturer
LLM**

Λ : behavior axes (frozen)
yes/no questions about one response

4 · Response collection

Target LLM
deployment
system prompt

replies
 $y^{P_1} \dots y^{P_N}$

5 · Response scoring

Judge LLM
blind to principal
one pass per level

6 · Comparing distributions

$P_1 \dots P_N$
one group per candidate

centroid $\bar{P}$
the pooled mixture

directional $\bar{I} = I(Z; X)$
between-group spread

common-mode ρ
overall firing rate

each half as excess over base
permutation null on prompt cells
deviation score per group

control calibration: the base model runs the identical pipeline;
a loyalty is a divergence the organism shows and its base does not

**secretly loyal?
to which principal?**

Differential Treatment Discovery



LLM component



frozen artifact



control calibration